Mahdieh Rojhannezhad, Ph.D.

Cancer Genomics Researcher | CRISPR Specialist | Molecular Geneticist | Genomic Data Analyst

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Languages: Persian (Native), English (Professional Working Proficiency)

Professional Profile

Innovative and research-driven geneticist with a Ph.D. in Genetics from Tarbiat Modares University and more than 5 years of multidisciplinary experience spanning cancer genomics, CRISPR/Cas9 gene editing, molecular diagnostics, genomic data interpretation, and scientific education. Specialized in regulatory genomics and enhancer biology with a strong focus on HER2-associated regulatory elements in breast cancer.

Experienced in genomic research, molecular and cytogenetic laboratory supervision, scientific mentoring, publication writing, peer review, and biotechnology development. Skilled in integrating molecular biology techniques with emerging computational genomics approaches, including whole exome sequencing (WES) interpretation, sequence analysis, variant analysis workflows, and bioinformatics tools.

Passionate about advancing precision medicine, functional genomics, cancer biology, and genomic interpretation through interdisciplinary research, biotechnology innovation, and scientific consulting.

Core Competencies

- Cancer Genomics & Functional Genomics
- CRISPR/Cas9 Gene Editing
- Regulatory Genomics & Enhancer Biology
- Whole Exome Sequencing (WES) Analysis
- Variant Interpretation & Genomic Data Analysis
- Molecular Diagnostics
- Breast Cancer Genetics
- Scientific Writing & Publication Development

- Translational Research
- Molecular & Cytogenetic Laboratory Supervision
- Bioinformatics & Sequence Analysis
- Primer and gRNA Design
- Cell Culture & Molecular Assays
- Research Mentorship & Scientific Communication
- Genomic Consulting & Research Support

Education

Ph.D. in Genetics

Tarbiat Modares University — Tehran, Iran

2016 – 2023

Dissertation focus: Functional and bioinformatic investigation of HER2-associated enhancer regions in breast cancer cell lines using CRISPR/Cas9 technology.

M.Sc. in Genetics

University of Tabriz — Tabriz, Iran

2009 – 2011

B.Sc. in Biology

University of Tabriz — Tabriz, Iran

2002 – 2007

Professional Experience

Supervisor — Molecular & Cytogenetic Departments

Watson Medical Genetics Laboratory — Tehran, Iran

January 2015 – January 2019

- Supervised molecular genetics and cytogenetics laboratory operations in diagnostic settings.
- Led teams of laboratory professionals and coordinated workflow optimization.
- Oversaw PCR, RT-PCR, qPCR, karyotyping, DNA/RNA extraction, and cytogenetic diagnostic procedures.
- Contributed to quality assurance, laboratory reporting accuracy, and diagnostic reliability.
- Participated in interpretation and technical evaluation of molecular diagnostic findings.
- Supported integration of molecular testing strategies in clinical genetics workflows.

Lecturer — Genetic Counseling

University of Rehabilitation Sciences & Social Welfare (Behzisti University) — Mashhad, Iran

October 2014 – January 2016

- Taught genetics and genetic counseling principles to undergraduate students.
- Mentored student research projects and scientific presentations.
- Delivered educational content related to molecular genetics, hereditary disorders, and clinical genomics.
- Strengthened scientific communication and interdisciplinary teaching capabilities.

Senior Specialist — Genetic Diagnostic Laboratories

Multiple Diagnostic Genetics Laboratories — Iran

2014 – 2023

- Performed molecular and cytogenetic laboratory analyses.
- Conducted DNA/RNA extraction, PCR-based assays, sequencing-related workflows, and cell culture procedures.
- Participated in genomic investigation and interpretation projects.
- Assisted in optimization of diagnostic and research protocols.
- Collaborated in research-oriented biotechnology and molecular development initiatives.

Research & Scientific Projects

HER2 Regulatory Enhancer Functional Analysis in Breast Cancer

- Conducted CRISPR/Cas9-mediated gene editing targeting HER2-associated enhancer regions.
- Investigated enhancer functionality and lncRNA GAS5 interactions in breast cancer cell lines.
- Performed functional genomics and regulatory element analysis in SKBR3, MCF7, and HEK293 cell lines.
- Combined molecular biology methods with bioinformatic analysis approaches.

Whole Exome Sequencing & Genomic Analysis

- Trained in clinical interpretation of WES datasets.
- Familiar with variant filtering strategies, genomic databases, sequence analysis tools, and genomic interpretation workflows.
- Interested in expanding expertise in computational genomics and precision medicine analytics.

Biotechnology Innovation Projects

- Developed improved DNA extraction methodology from paraffin-embedded tissue samples.
- Participated in development of cell culture media formulations.
- Contributed to nucleotide size standard development projects.
- Pursued interdisciplinary exploration in drug design and genomic biotechnology.

Publications

Scientific Reports (Springer Nature), 2023

Rojhannezhad M, Mowla SJ.
Functional analysis of a HER2-associated expressed enhancer in breast cancer cells.

Journal of Advanced Immunopharmacology, 2024

Rojhannezhad M, Mowla SJ.
Gene Editing for Unraveling the Regulatory Role of a HER2-Associated Enhancer with lncRNA GAS5 and Related Genes in Breast Cancer Cells.

Zahedan Journal of Research in Medical Sciences, 2015

Rojhannezhad M, Hoseinpourfeizi MA, Pouladi N.
The association between Survivin -31 G/C promoter polymorphism and breast cancer risk.

Sarem Journal of Medical Research, 2023

Rojhannezhad M.
Functional and Bioinformatic Investigation of the Regulatory Region of HER2-Associated Enhancer GH17J039694 (HER2-Enhancer1) Using CRISPR/Cas9 Technology.

Reviewer & Academic Activities

Reviewer

Journal of Cancer and Tumor International

- Review manuscripts related to oncology, cancer biology, molecular genetics, and biomedical sciences.
- Evaluate scientific rigor, methodological quality, and publication relevance.

Technical Skills

Molecular Biology & Laboratory Techniques

- PCR, RT-PCR, qPCR
- CRISPR/Cas9 Gene Editing
- DNA & RNA Extraction
- Gel Electrophoresis
- Cloning & Transfection
- Human & Animal Cell Culture
- Amniotic Fluid Cell Culture
- Cytogenetics & Karyotyping
- Genetic Counseling
- Protein & RNA Analysis
- MTT & Scratch Assays

Bioinformatics & Computational Genomics

- Whole Exome Sequencing (WES) Analysis
- Variant Interpretation Workflows
- Primer & gRNA Design
- Sequence Analysis Tools
- Cytoscape
- Bioinformatics Databases
- Drug Design Databases
- Genomic Data Interpretation
- Regulatory Genomics Analysis

Software & Platforms

- BioRender
- ASI Karyotyping Software
- Adobe Illustrator
- Microsoft Office Suite
- AI-Assisted Research Tools

Programming & Computing

- Python (Basic)
- R Programming (Basic)
- Linux Environment (Basic)

Certifications & Professional Development

- Clinical Analysis of Whole Exome Sequencing Data
- Next Generation Sequencing (NGS) Workshop
- Cancer Systems Biology
- Protein Bioinformatics Workshop
- Cytoscape Software Workshop
- Genetic Engineering Workshops
- Gene Expression Analysis Training Course
- Molecular Biology Workshops
- Clinical Cytogenetics Workshop
- Non-Coding RNA Training Course
- Stem Cell Systems Biology
- Royan International Summer School

- First International Congress on Biomedicine
- Genetic Counseling Workshops

Honors & Achievements

- Ranked 8th nationally in Iran's Ph.D. Entrance Examination for Medical Genetics (2014)
- Ranked 13th nationally in Iran's Ph.D. Entrance Examination for Molecular Genetics (2017)
- Developed enhanced DNA extraction methodology for paraffin-embedded tissue samples
- Participated in development of advanced cell culture media and nucleotide size standards
- Contributed to interdisciplinary biotechnology innovation projects

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Scientific Interests

- Cancer Genomics
- Functional Genomics
- Precision Medicine
- Regulatory Elements & Enhancer Biology
- CRISPR Technologies
- Computational Genomics
- Bioinformatics
- Molecular Oncology
- Drug Design
- Translational Biotechnology
- Neuroscience

Additional Information

Hobbies & Personal Interests

Applied biotechnology innovation, scientific prototyping, computational biology exploration, biomaterials research, martial arts (Karate), swimming, scientific reading, and interdisciplinary biomedical development.

References

Available upon request.